Seminar 3

Two routes through the consumer-theory square

Conventions. Quantities are nonnegative and $p_1, p_2, m > 0$. Targets satisfy $\bar{u} > 0$ in problem 1 and $\bar{u} > 1$ in problem 2 unless stated otherwise. Write X, H for demand correspondences and x, h when demand is unique. Use derivative identities where their derivatives exist.

Warm-up: true or false

Assume continuous, locally nonsatiated preferences and attainable targets. Explain each verdict briefly.

1. At fixed prices, a higher utility target cannot reduce minimum expenditure.
2. Raising a price must reduce the expenditure needed to attain a fixed target, because the consumer can substitute away from that good.
3. Multiplying every price by the same positive number leaves the Hicksian demand correspondence unchanged.
4. If two bundles minimise expenditure for the same target, their midpoint must also minimise expenditure.
5. At matching income and utility, Marshallian and Hicksian demand correspondences coincide, even if the optimum is not unique.
6. Wherever both demands are differentiable, their own-price derivatives must coincide at matching parameters.

1. A little of everything

A consumer enjoys two divisible activities:

$$u(x_1, x_2) = \sqrt{x_1} + 2\sqrt{x_2}.$$

(a) Establish existence, uniqueness and budget exhaustion. Why must both optimal quantities be strictly positive?

(b) Solve utility maximisation directly: find $x_1(p, m)$, $x_2(p, m)$ and $v(p, m)$ for general prices and income.

(c) Solve expenditure minimisation directly: find $h_1(p, \bar{u})$, $h_2(p, \bar{u})$ and $e(p, \bar{u})$. Try $z_i = \sqrt{x_i}$.

(d) Recover x using Roy and h using Shephard. Verify

$$e(p, v(p, m)) = m, \qquad v(p, e(p, \bar{u})) = \bar{u}, \qquad h(p, v(p, m)) = x(p, m).$$

(e) At $p = (1, 4)$ and $m = 8$, find x and v. Calculate x_{1,p_1}, $x_{1,m}$ and h_{1,p_1} at the matching *fixed* target. Check the own-price Slutsky equation numerically.

(f) Compute the full Slutsky matrix from Marshallian demand:

$$S_{ij} = \frac{\partial x_i}{\partial p_j} + x_j \frac{\partial x_i}{\partial m}.$$

Check it against $D_p h$ at the matching target. Verify symmetry, $Sp = 0$ and negative semidefiniteness. Give its value at the point in (e).

(g) Generalise. For an n-good consumer with continuous LNS utility, positive prices and an attainable target above $u(0)$, suppose demands are single-valued and differentiable and e is C^2 in prices. At matching parameters, prove that S is symmetric, $Sp = 0$ and $z^\top S z \leq 0$. Use Shephard and homogeneity; derive price concavity of e from its minimisation definition.

2. Two escape routes and a head start

The second consumer has utility

$$u(x_1, x_2) = \max\{x_1 + 1, \sqrt{x_2}\}.$$

(a) Check continuity, local nonsatiation and convexity of preferences. Draw $\{x \geq 0 : u(x) \geq 2\}$. What is $u(0)$, and why do we work with targets $\bar{u} > 1$ for the main duality calculations?

(b) Find $v(p, m)$ and the entire $X(p, m)$ by comparing the two corner routes. At $p = (9/2, 1)$, find every income range in which each route is optimal, including both switching boundaries.

(c) Find $e(p, \bar{u})$ and the entire $H(p, \bar{u})$ for $\bar{u} > 1$. At $p = (9/2, 1)$, find every target range and both switching targets. What happens for targets $\bar{u} \leq 1$?

(d) Apply Roy and Shephard in each strict regime. Explain what prevents a single gradient from selecting a bundle at a switching boundary.

(e) Verify both inverse-value identities and demand-set equality at matching parameters. At the specified prices, match the two income boundaries to the two target boundaries.

(f) At $p = (9/2, 1)$ and $m = 9/4$, list all solutions to both corresponding problems. Is the midpoint also optimal? Check the other boundary, $m = 9$.

(g) *Extension.* Set $p = (4, 1)$ and $m = 4$. Does the derivative v_m exist at this tie? Does that alone let you use Roy's identity?